

README — CMP2204 Term Project: Peer-to-Peer Chat Application

This project is a Peer to Peer chat application based on Python. The system provides real-time encrypted and unencrypted messaging between users across the same Local Area Network (LAN) using UDP for peer discovery and TCP for messaging.

Dependencies

Ensure the following Python libraries are installed:

1- pip install pyDes

2- pip install PyQt5

Terminal Version (Modular)

Terminal 1

```
python3 UDPclient.py # Service Announcer
```

Terminal 2

```
python3 UDPserver.py # Peer Discovery
```

Terminal 3

```
python3 TCPserver.py # Chat Responder
```

Terminal 4

```
python3 TCPclient.py # Chat Initiator
```

All-in-One Console

```
python3 chatui.py
```

Features

- **Peer Discovery & Service Announcement:**
 - Users broadcast their presence every 8 seconds via UDP on 192.168.1.255:6000.
 - Peer discovery listens and stores active users in a dictionary with timestamps.
- **User Availability Status:**
 - Users online in the last 10 seconds are marked as (Online), others as (Away).
 - Only users seen within the last 15 minutes are displayed.
- **Encrypted and Unencrypted Chat:**
 - Diffie-Hellman key exchange implemented.
 - Encrypted messages use pyDes and base64 encoding.
 - Unencrypted messages are sent in plaintext JSON.
- **Chat Logging:**
 - All messages are logged with timestamps, sender/receiver info, and message content.
- **User Interface:**
 - Terminal-based interactive menu with options:
 - Users – show available users.
 - Chat – initiate chat with secure/unsecure option.
 - History – view complete chat history.(You can also clear the chat history.)
 - Graphical User Interface:
 - Peer List Panel: Displays all discovered users with online/away/offline status
 - Chat Window: Shows the conversation history with secure/unsecure messages.
 - Message Input Field: Allows users to type and send messages.
 - Secure Checkbox: Enables sending encrypted messages using Diffie-Hellman key exchange and pyDes.
 - Buttons
 - Send: Sends the message.
 - View Log: Displays the entire chat history in the chat window.
 - Clear Log: Deletes the entire chat history.
 - Users are prompted to enter a username at launch.
 - A connection is automatically established when a peer is selected from list.
- **Username Prompt:**
 - Upon launching either the terminal version (UDPclient.py, TCPserver.py) or the all-in-one GUI version (chatui.py), the application will prompt the user to enter a username.
 - This username will be included in UDP broadcasts for peer discovery and used to identify the sender/receiver in chat messages.

Known Limitations

- The system is designed for and tested only on LANs with subnet 192.168.1.X.
- If multiple users have the same username, ambiguity may occur in chat targeting.
- Message size is limited by TCP buffer size (~1024 bytes).
- GUI is terminal-based only.

Developers

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